

1. What answers get confused with each other most easily? What kinds of mistakes does this guesser make?

In general, the model struggles when it is difficult to discern what the format of the answer will be. For example, one question is “A composer from this century names a cadence in which the leading tone goes down to the submediant before resolving up to the tonic ... For 10 points, name this century during which isorhythmic motets were written by Ars nova composers such as Philippe de Vitry, whose music was admired by his contemporary, Petrarch.” The guess was “Pedro\_de\_Alcântara,\_Prince\_of\_Grão-Pará”, which likely matched the low frequency words in the question but was wrong because the type of answer given was incorrect. Conversely, another consistent error type is when the model provides highly-specific, low frequency answers in place of relatively simple, higher frequency answers. For example, “Refractive\_index” in place of “Ocean” or “Continuous\_function” in place of “Symmetry”.

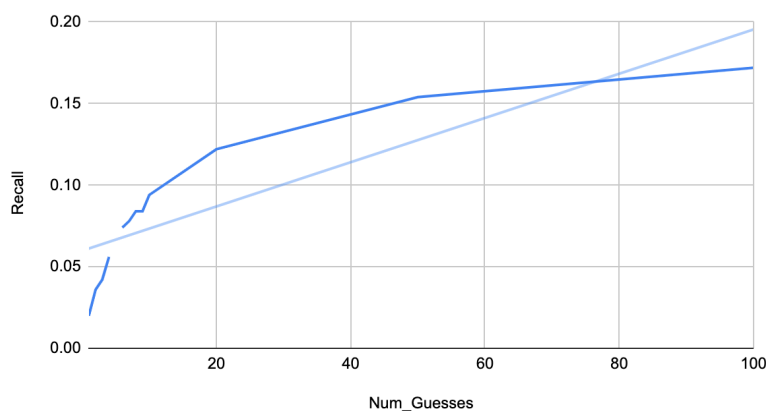
2. How does this guesser compare to GPT? (Remember that the cached guesser from the feature engineering homework came from GPT3, so you could either use your old code or adapt with multiple guessers here!)

The guesser here, with my implementation, does much worse than GPT. Using the eval script on the cached guesser, I get a precision and recall of .8600. By contrast, with 100 for num\_guesses, I get a Recall of 0.172 and Precision of .200.

3. Compute recall as you increase the number of guesses (i.e. num\_guesses).

Below, the value for -- limit was kept consistent for training.

Recall vs. Num\_Guesses



The data for this chart is linked here:

[https://docs.google.com/spreadsheets/d/1LcwQkyNYlvgtvJPbEPrOhPex4F\\_8RwXDxp8rNfX1-Co/edit?usp=sharing](https://docs.google.com/spreadsheets/d/1LcwQkyNYlvgtvJPbEPrOhPex4F_8RwXDxp8rNfX1-Co/edit?usp=sharing)